

Ranking Reversal Theory Under Candidate-Set Shift in GRN Benchmarking

Anonymous Author(s)

ABSTRACT

When method A scores higher than method B on a gene regulatory network (GRN) benchmark, does the rank still hold if we change the evaluation protocol — candidate edges or reference network? The answer hinges on the *margin* between the methods’ scores, and we give a closed-form rule for when the rank is safe. Across four protocol axes on six methods and three single-cell tissues we find: candidate-set changes flip 16% of method pairs (cluster-bootstrap 95% CI 7–27%); cross-tissue, 19% (9–31%); reference-network, 32% (20–43%); gene-symbol mapping, none of 165. The popular “base-rate $\times$ discrimination” decomposition has two pieces the algebra forces (no data information); the data-carrying piece is the relative *magnitude* of the two terms, which we find large and significant (rank-biserial $r = 0.67$, cluster-permutation $p < 2 \times 10^{-4}$). A single leaderboard number is not biological evidence on its own: reports should include protocol stability, margin regime, and metric.

CCS CONCEPTS

• **Applied computing** $\rightarrow$ **Computational genomics**; • **Theory of computation** $\rightarrow$ *Mathematical optimization*.

KEYWORDS

gene regulatory networks, evaluation bias, ranking stability, benchmark lottery, cluster bootstrap

1 INTRODUCTION

GRN benchmark leaderboards are read as evidence for biological plausibility [1, 2], but behind every score is a chain of protocol choices — candidate edges, reference network, gene-symbol mapping, tissue — rarely reported or controlled [3–6]. The problem is familiar in ML: BLEU re-ranks under different tokenizers [11]; “benchmark lottery” effects swap leaderboards across dataset choices [12–14]. When two methods’ ranks flip with the protocol, the biological consequence flips too: which regulator gets validated, which model is called credible. We need a way to say when a rank is robust.

Contributions. (1) A closed-form “safety margin” per method pair (§2). (2) A reading of the base-rate-vs.-discrimination decomposition separating sign claims (forced by algebra) from magnitude claims (an actual data fact) (§4.2). (3) An audit across four protocol axes with cluster-bootstrap intervals, three nulls, a margin sweep, a learned classifier (honest negative), and a metric-robustness check (§4).

2 THEORY

Setup. Let $\mathcal{M}_m(S, \pi, R)$ be a benchmark score (such as AUPR) for method m under candidate set S , gene-symbol mapping π , and reference network R . The *margin* between methods A and B is the score gap $\Delta = \mathcal{M}_A - \mathcal{M}_B$, and a *ranking reversal* between two

protocol settings is the event $\Delta_1 \Delta_2 < 0$ — the gap flips sign. Writing $\Delta = b \cdot g$ with b the positive-class base rate and $g := \Delta/b$ the base-rate-normalized gap, the standard “base-rate $\times$ discrimination” decomposition follows directly from the product rule:

$$\Delta_2 - \Delta_1 = \underbrace{(b_2 - b_1)g_1}_{\text{base-rate term}} + \underbrace{b_2(g_2 - g_1)}_{\text{calibration term}}. \quad (1)$$

The current margin $|\Delta_1|$ is the “cushion”: for the rank to flip, the shift must first eat through it and overshoot in the opposite direction (Prop. 1).

PROPOSITION 1 (MINIMUM-SHIFT BOUND). *If every shift in a family satisfies $|\delta\Delta| \leq B$, then no method pair with margin $|\Delta_1| > B$ can reverse. More tightly, a reversal of a pair with margin $|\Delta_1|$ requires $|b_2 - b_1||g_1| + b_2|g_2 - g_1| \geq |\Delta_1|$. In particular, if discrimination is shift-stable ($|g_2 - g_1| \leq \epsilon_g$), any reversing base-rate change must satisfy $|b_2 - b_1| \geq (|\Delta_1| - b_2\epsilon_g)/|g_1|$ (informative when $b_2\epsilon_g < |\Delta_1|$).*

Sign counts from Eq. 1 are forced by algebra. Under positive base rates, holding g fixed gives $(b_2/b_1)\Delta_1^2 > 0$ (a base-rate change alone never flips the sign); holding b fixed gives $(b_1/b_2)\Delta_1\Delta_2 < 0$ on every reversal by definition. So the “0/22 vs. 22/22” counts reported by earlier versions [6] restate the definition — the only data-bearing piece is the magnitude comparison (§4.2).

3 METHODS

Data. Published benchmark medians of AUPR, AUROC, and F1 across three Tabula Sapiens tissues [7] (kidney, lung, immune) $\times$ three candidate-set definitions, on six methods: GENIE3 [5], GRNBOOST2, PEARSON, SPEARMAN, SCGPT_ATTENTION [1], RANDOM. Reference shifts on the immune slice with three reference networks (HPN_DREAM, BEELINE_GSD [3], a DOROTHEA [8]/TRRUST [9] union); mapping shifts on a probe-prior subset over four symbol-resolution policies (OMNIPATH [10] included).

Analyses. (i) per-pair reversal indicator $1\{\Delta_1\Delta_2 < 0\}$; (ii) **cluster-bootstrap 95% confidence intervals** treating each method pair as one cluster [15] (Wilson intervals would shrink the CI artificially by assuming independence); (iii) a **cluster-restricted permutation Mann-Whitney test** on the magnitude-ratio comparison, with 5,000 within-cluster permutations of the reversal label; (iv) per-cell Benjamini–Hochberg false discovery rate; (v) a margin-magnitude sweep over a threshold τ ; (vi) three null models: random method-label permutation, Gaussian noise ($\sigma = 10\%$ of within-cell median margin), within-cell rank permutation; (vii) leave-one-tissue-out reversal prediction: quantile heuristic vs. logistic regression on $|\Delta_1|, b_1, b_2 - b_1, |g_1|$; (viii) metric robustness across AUPR/AUROC/F1. All seeds fixed at 42; decomposition closure verified below 10^{-9} .

Table 1: Headline reversal rates by protocol axis with cluster-bootstrap 95% CIs. Tissue is reported three ways because they answer different questions (see §4.1).

Axis	n	k	Rate	95% CI
Candidate-set shift	135	22	0.16	[0.07, 0.27]
Tissue shift (cand-cond.)	135	26	0.19	[0.09, 0.31]
Tissue shift (dedup, any cand.)	45	17	0.38	[0.18, 0.58]
Tissue shift (dedup, all cand.)	45	1	0.02	[0.00, 0.07]
Reference shift (immune)	106	34	0.32	[0.20, 0.43]
Mapping-policy shift	165	0	0.00	[0.00, 0.00]

4 RESULTS

4.1 Candidate-set and tissue axes

The marginal 16% candidate-shift rate is driven by the immune tissue and by pairs of methods that were already close (§4.3). After per-cell Benjamini–Hochberg correction, *none* of the high-rate cells highlighted in the workshop precursor [6] (e.g. the immune all-pairs to TF-source-target shift at 6/15) survive $q < 0.05$. What does survive is the opposite: cells that are significantly *more stable* than chance (e.g. the kidney TF-source to TF-source-target shift at 0/15, $q = 5 \times 10^{-4}$).

For tissue shifts, the same method pair appears in three candidate-set cells, so counting them separately inflates the signal. When we deduplicate to one row per (method-pair, tissue-pair) and ask “does this pair flip under at least one candidate set?”, the rate is 0.38 (17/45); asking “does it flip under *all* candidate sets?” gives 0.02 (1/45). The honest single-number summary is the “flips at least once” rate, 0.38.

4.2 Magnitude decomposition

In all 22 candidate-shift reversal rows, the discrimination (“calibration”) term is larger in magnitude than the base-rate term (22/22 vs. 56/113 in non-reversal rows). The mean ratio $|\text{cal}|/|\text{base-rate}|$ is 1.54 in reversals versus 0.83 in non-reversals.

A naive Mann–Whitney test would give $p = 3.6 \times 10^{-7}$, but the 135 rows are not independent — they come from only 15 unique method-pair clusters (and only 7 of those clusters contain any reversal). To respect that dependence we ran a *cluster-restricted* permutation Mann–Whitney test that shuffles the reversal labels only *within* each method-pair cluster (5,000 permutations). The observed test statistic $U = 2075$ sits above the null 97.5% quantile ([1426, 1907]), with rank-biserial effect size $r = 0.67$ (large) and permutation $p < 2 \times 10^{-4}$ (no null draw exceeded the observed U). This cluster-aware p is the one we report; the iid p above is too optimistic.

4.3 Margin regime, nulls, screening, metric

Margin sweep. Discarding method pairs whose initial gap is below the 75th percentile, candidate-shift reversals vanish (0/34); reference-shift reversals do not (rate stays in [0.22, 0.38]). So candidate-shift instability is a *near-tie phenomenon*, while reference-shift instability survives clear gaps.

Three nulls. The observed 0.16 sits well below all three: random method-label permutation gives null mean 0.50 ([0.37, 0.61]); within-cell rank permutation, the same; Gaussian noise at $\sigma = 10\%$ of typical gap lifts the rate to 0.29 ([0.21, 0.37]). Signal is real and robust to $\sim 10\%$ -of-margin noise.

Can we predict which pairs flip? Not at this sample size. The quantile heuristic peaks at $F1 = 0.35$ (precision 0.24, recall 0.64, PR-AUC 0.18); a learned logistic regression on $|\Delta_1|, b_1, b_2 - b_1, |g_1|$ does *worse* (PR-AUC 0.13, below the baseline 0.16). Reversal is a population-level phenomenon here, not per-pair predictable from these features.

Metric choice changes the answer. AUPR flips 4/45 tissue pairs (0.09), F1 the same 4/45, AUROC 0/40 (Wilson upper 0.09). AUROC’s invariance to any monotone re-scoring preserves cross-tissue rank similarity; AUPR’s base-rate sensitivity does not. *Which metric you pick can swing the conclusion entirely.*

Worked example. A kidney pair (SCGPT_ATTENTION vs. SPEARMAN, TF-source to TF-source-target shift) has gap $|\Delta_1| = 2.85 \times 10^{-2}$ — safely above any shift family in our data. An immune pair (PEARSON vs. SPEARMAN, all-pairs to TF-source-target) has gap 8.24×10^{-7} — essentially zero, and indeed one of the 22 reversals, but a coin flip on noise. The 75th-percentile margin cutoff (§4.3) excises pairs like the second, dropping the rate to 0/34.

Limitations. Summary medians only (no replicate noise); one tissue for the reference axis; one slice for mapping; Prop. 1 is necessary, not sufficient.

5 DISCUSSION AND RECOMMENDATIONS

Three patterns: (i) when candidate-axis reversals happen, the calibration (discrimination) term is consistently larger than the base-rate term in magnitude — a real data finding, not a tautology; (ii) candidate-shift reversals disappear once methods are clearly separated, reference-shift ones do not; (iii) metric choice alone can swing the conclusion [13]. Four practical recommendations: cluster-bootstrap CIs over method pairs (not Wilson); state the margin regime; report at least two metrics (AUPR + AUROC) per rank claim; retain the calibration-vs-base-rate magnitude ratio (not the sign counts, which are forced by the algebra).

Reproducibility. <https://anonymous.4open.science/r/grn-ranking-reversal-theory-DB8E/>

REFERENCES

- [1] H. Cui, C. Wang, H. Maan, et al. scGPT: toward building a foundation model for single-cell multi-omics using generative AI. *Nat. Methods* 21(8):1470–1480, 2024.
- [2] C. V. Theodoris et al. Transfer learning enables predictions in network biology from large-scale single-cell atlases. *Nature* 618:616–624, 2023.
- [3] A. Pratapa, A. P. Jaliha, J. N. Law, A. Bharadwaj, T. M. Murali. Benchmarking algorithms for gene regulatory network inference from single-cell transcriptomic data. *Nat. Methods* 17:147–154, 2020.
- [4] S. Aibar, C. B. González-Blas, T. Moerman, et al. SCENIC: single-cell regulatory network inference and clustering. *Nat. Methods* 14:1083–1086, 2017.
- [5] V. A. Huynh-Thu, A. Irthum, L. Wehenkel, P. Geurts. Inferring regulatory networks from expression data using tree-based methods. *PLoS ONE* 5(9):e12776, 2010.
- [6] Anonymous authors. Evaluation bias and symbol mapping in GRN benchmarks. Workshop manuscript, 2025. (Author identity withheld for double-blind review.)
- [7] Tabula Sapiens Consortium. The Tabula Sapiens: a multiple-organ single-cell transcriptomic atlas of humans. *Science* 376:eab4896, 2022.
- [8] C. H. Holland, J. Tanevski, J. Perales-Patón, et al. DoRothEA as a resource of transcription factor activities from gene expression data. *Nat. Commun.* 11:589, 2020.
- [9] H. Han, J.-W. Cho, S. Lee, et al. TRUST v2: an expanded reference database of human and mouse transcriptional regulatory interactions. *Nucleic Acids Res.* 46(D1):D380–D386, 2018.
- [10] D. Türe, A. Valdeolivas, L. Gul, et al. Integrated intra- and intercellular signaling knowledge with OmniPath. *Nat. Methods* 18:1303–1316, 2021.
- [11] M. Post. A call for clarity in reporting BLEU scores. *WMT*, 2018.
- [12] M. Dehghani, Y. Tay, A. A. Gritsenko, et al. The benchmark lottery. *arXiv:2107.07002*, 2021.
- [13] J. Demšar. Statistical comparisons of classifiers over multiple data sets. *JMLR* 7:1–30, 2006.
- [14] D. E. Critchlow. *Metric Methods for Analyzing Partially Ranked Data*. LNS 34, Springer, 1985.
- [15] A. C. Cameron, D. L. Miller. A practitioner’s guide to cluster-robust inference. *J. Hum. Res.* 50:317–372, 2015.